

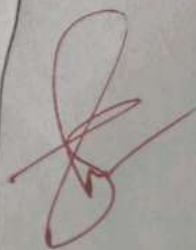
5.

Present grayscale image representation, identifying practical limitations, analyzing them using examples and tools, and ensuring clear and Professional communication.

Technical content accuracy (30)	Problem Identification (25)	Use of Tools (20)	Communication (15)	Professionalism (10)
25	22	20	15	60

Total
(100)

92





 GPS Map Camera

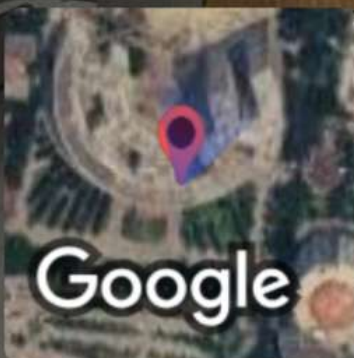
Mevalurkuppam, Tamil Nadu, India



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602105, India

Lat 13.02587° Long 80.016683°

Thursday, 16/07/2026 01:44 PM GMT +05:30





SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Grayscale Image Representation In Computer Vision

ITA0506 – Computer Vision

Guided By:

Presented By:

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INTRODUCTION



What is a Grayscale Image?

- A grayscale image contains only shades of **black, white, and Gray**.
- Each pixel has a single intensity value.
- The pixel intensity ranges from **0 to 255**.
- It removes color information and keeps only brightness.

Intensity Values

- ✓ 0 = Black
- ✓ 255 = White
- ✓ 1 – 254 = Shades of Gray



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IMPORTANCE OF GRAYSCALE IMAGES



Why do we use Grayscale Images?

- Reduces image complexity.
- Uses less memory.
- Faster image processing.
- Improves edge detection.
- Simplifies Computer Vision algorithms.

WORKING PRINCIPLE

☐ Workflow

Color Image (RGB)



Read Image using OpenCV



Convert RGB to Grayscale



Display Grayscale Image



Use for Computer Vision Applications



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APPLICATIONS OF GRAYSCALE IMAGES



- Face Recognition
- Medical Imaging
- OCR (Text Recognition)
- Fingerprint Recognition
- CCTV Surveillance
- Autonomous Vehicles
- Industrial Inspection



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ADVANTAGES & LIMITATIONS



Advantages

- Faster processing
- Less memory usage
- Easy image analysis
- Better edge detection
- Simple implementation

Limitations

- No color information
- Difficult to distinguish objects with similar intensity
- Less suitable when color is important



IMPLEMENTATION



Software Used

- Python OpenCV
- VS Code
- Windows Operating System

Algorithm

- Import OpenCV library.
- Read the input image.
- Convert the image into grayscale.
- Display the original image.
- Display the grayscale image.



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PYTHON CODE



```
File Edit Selection View Go Run Terminal Help
Grayscale_project

EXPLORER
GRAYSCALE_PROJECT
  Grayscale.py
  image.jpg

Grayscale.py
1 import cv2
2
3 # Read the image
4 img = cv2.imread("image.jpg")
5
6 # Check if the image exists
7 if img is None:
8     print("Image not found!")
9 else:
10     # Convert to grayscale
11     gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)
12
13     # Show original image
14     cv2.imshow("Original Image", img)
15
16     # Show grayscale image
17     cv2.imshow("Grayscale Image", gray)
18
19     # Wait until a key is pressed
20     cv2.waitKey(0)
21
22     # Close all windows
23     cv2.destroyAllWindows()
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

PS C:\Users\Rama Krishna K\Desktop\Grayscale_project> python Grayscale.py

PS C:\Users\Rama Krishna K\Desktop\Grayscale_project>

powerShell
powerShell
powerShell
powerShell

Tip: Try the Plan agent to research and plan before implementing changes.

+ Grayscale.py

Describe what to build

+ | | Auto |

Local | Default Approvals

Ln 23, Col 28 Spaces: 4 UTF-8 CRLF {} Python

CONCLUSION

- Grayscale images simplify image processing.
- They reduce memory usage and improve speed.
- They are widely used in Computer Vision applications.
- However, they cannot preserve color information.